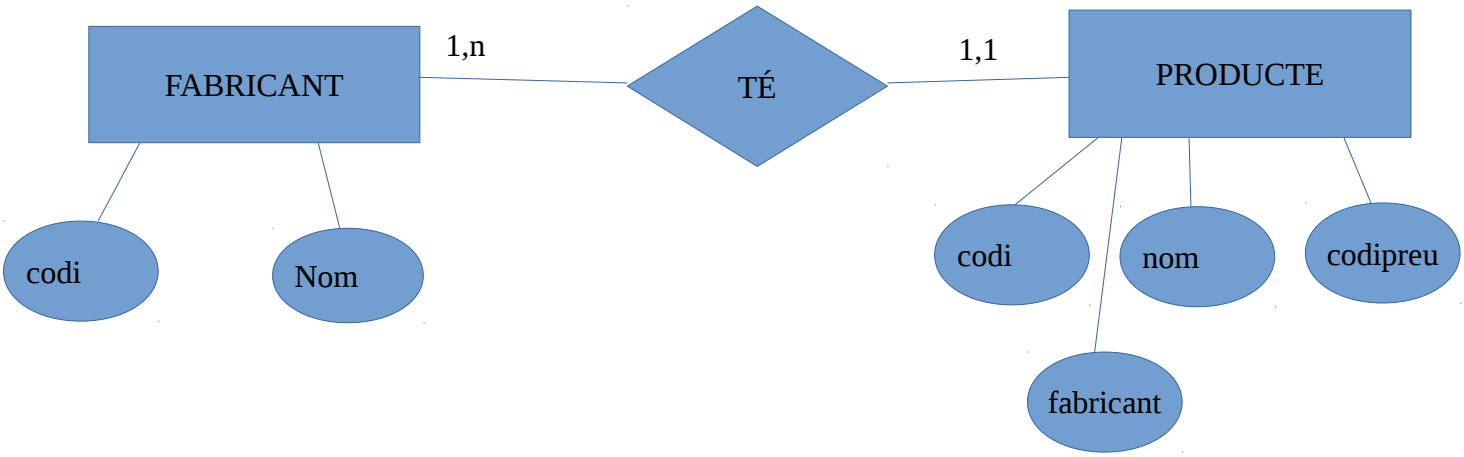
Tasca 003 Llenguatge SQL.

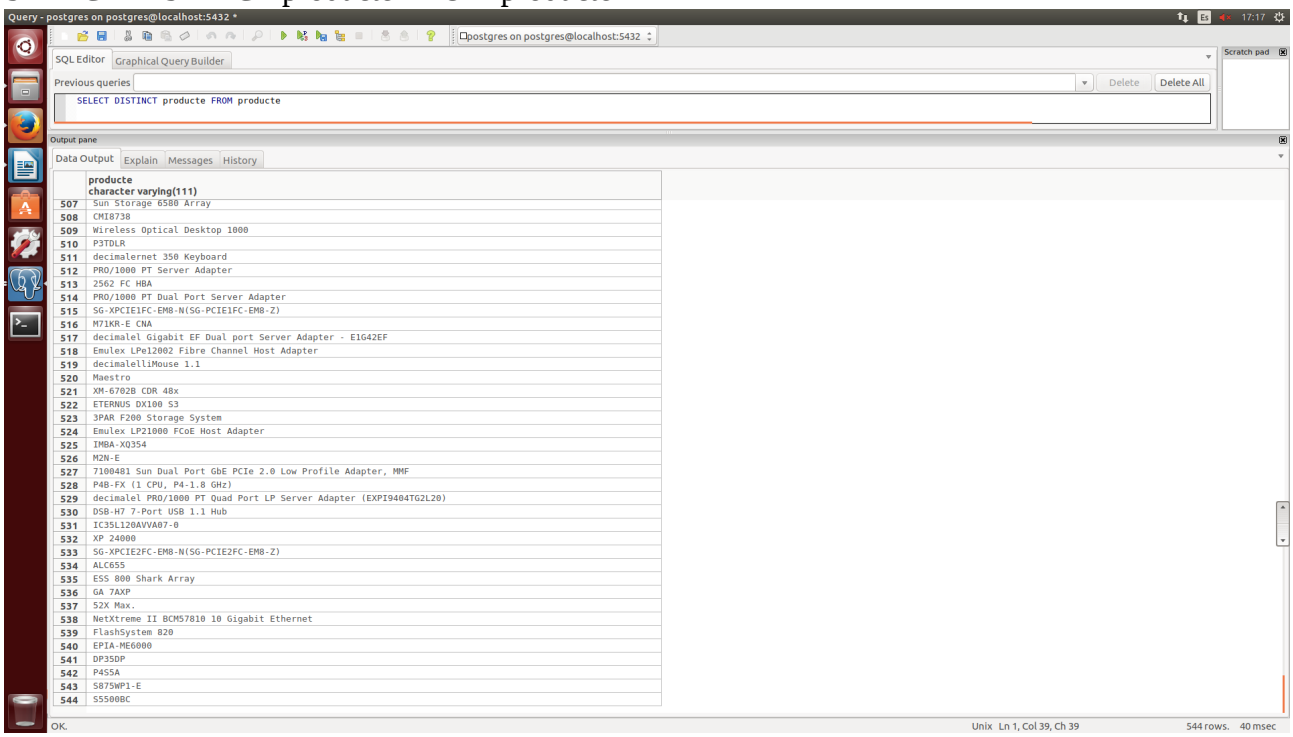
La botiga d'Informàtica

Fixat en el següent diagrama E/R i confecciona les consultes que et demanem a continuació.



1. Obté el nom dels productes de la botiga.

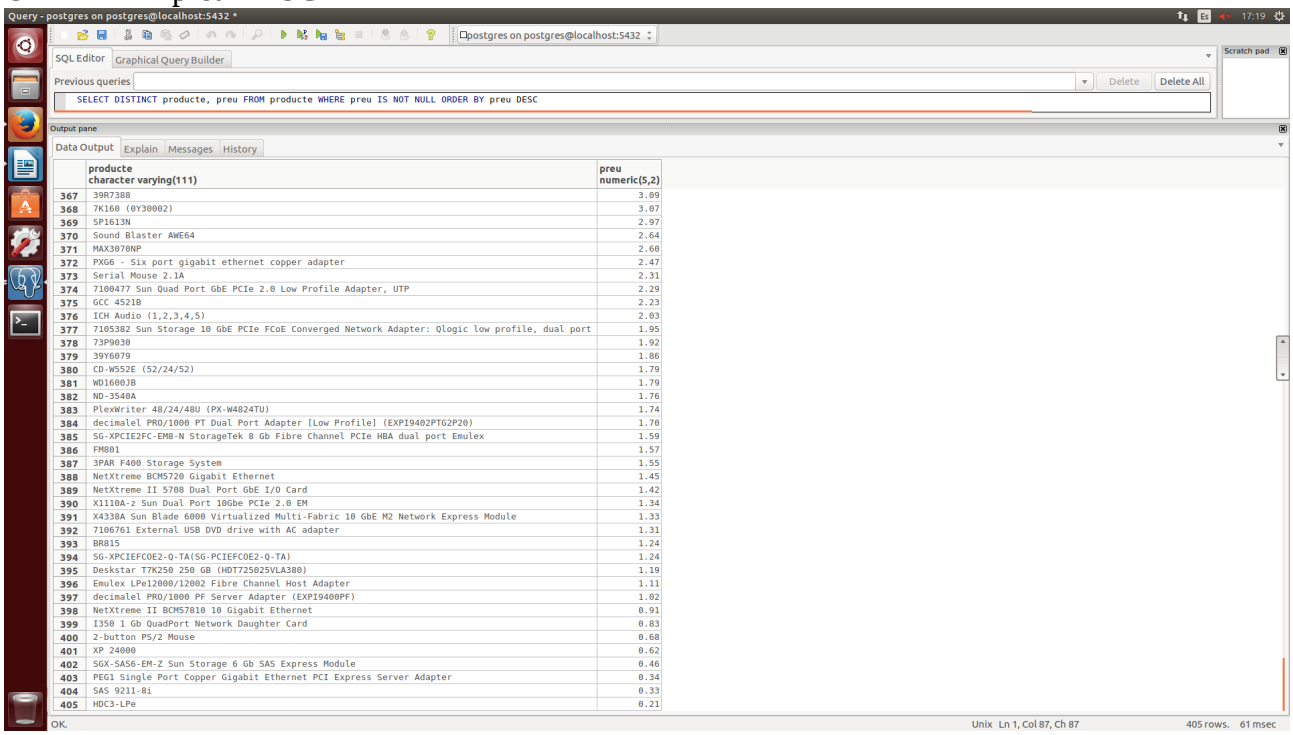
SELECT DISTINCT producte FROM producte



Tasca 003 Llenguatge SQL.

2. Nom i preu dels productes de la botiga ordenats des del més car al més barat.

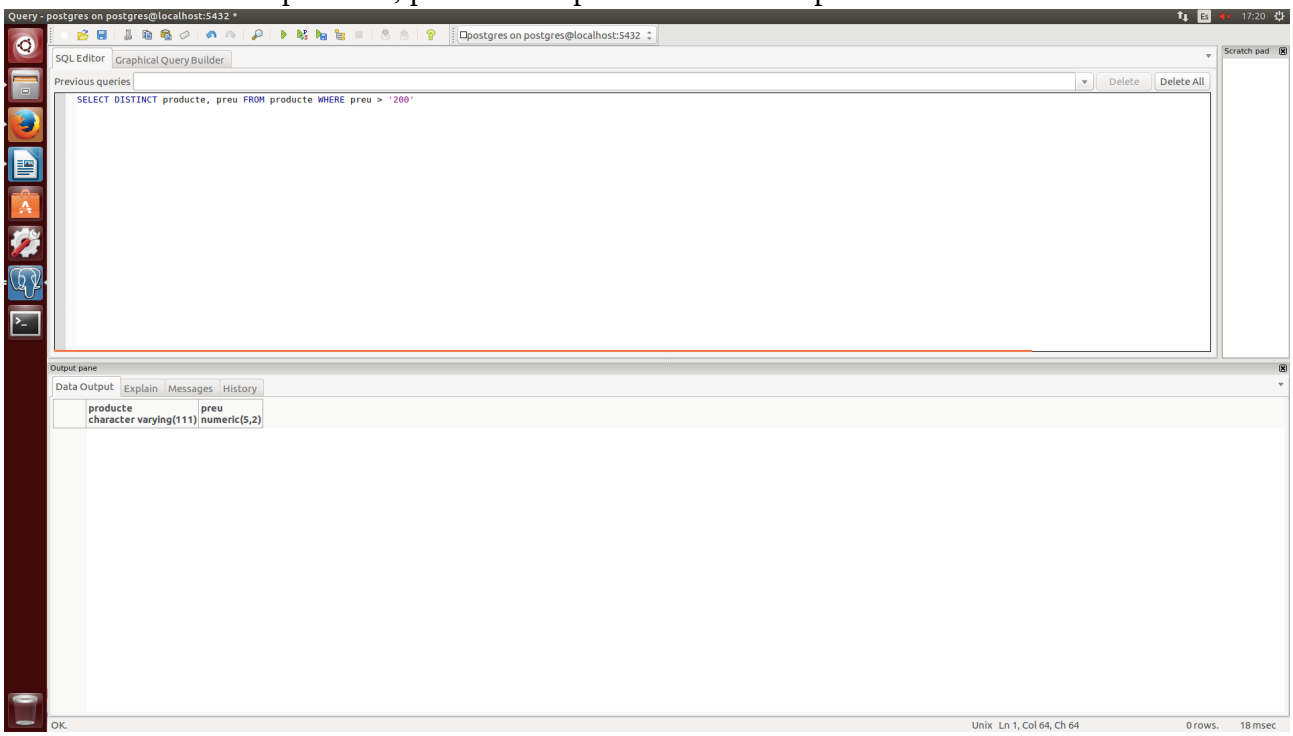
SELECT DISTINCT producte, preu FROM producte WHERE preu IS NOT NULL
ORDER BY preu DESC



	producte character varying(111)	preu numeric(5,2)
367	39R7388	3.09
368	7K160 (0Y30002)	3.07
369	SP1613N	2.97
370	Sound Blaster AWE64	2.64
371	MAX3070NP	2.60
372	PXG6 - Six port gigabit ethernet copper adapter	2.47
373	Serial Mouse 2.1A	2.31
374	7100477 Sun Quad Port GbE PCIe 2.0 Low Profile Adapter, UTP	2.29
375	GCC 4521B	2.23
376	ICH Audio (1,2,3,4,5)	2.03
377	7105382 Sun Storage 10 GbE PCIe FCoE Converged Network Adapter: Qlogic low profile, dual port	1.95
378	73P9030	1.92
379	39Y6079	1.86
380	CD-W552E (52/24/52)	1.79
381	WD1600JB	1.79
382	ND-3540A	1.76
383	PlexWriter 48/24/48U (PX-W4824TU)	1.74
384	decimalel PRO/1000 PT Dual Port Adapter [Low Profile] (EXPI9402PTG2P20)	1.70
385	SG-XPCIE2FC-EM8-N StorageTek 8 Gb Fibre Channel PCIe HBA dual port Emulex	1.59
386	FM801	1.57
387	3PAR F400 Storage System	1.55
388	NetXtreme BCM5720 Gigabit Ethernet	1.45
389	NetXtreme II 5708 Dual Port GbE I/O Card	1.42
390	X1110A-z Sun Dual Port 10Gbe PCIe 2.0 EM	1.34
391	X4338A Sun Blade 6000 Virtualized Multi-Fabric 10 GbE M2 Network Express Module	1.33
392	7106761 External USB DVD drive with AC adapter	1.31
393	BR815	1.24
394	SG-XPCIEFCOE2-Q-TA(SG-PCIEFCOE2-Q-TA)	1.24
395	Deskstar T7K250 250 GB (HDT725025VLA380)	1.19
396	Emulex LPe12000/12002 Fibre Channel Host Adapter	1.11
397	decimalel PRO/1000 PF Server Adapter (EXPI9400PF)	1.02
398	NetXtreme II BCM57810 10 Gigabit Ethernet	0.91
399	I350 1 Gb QuadPort Network Daughter Card	0.83
400	2-button PS/2 Mouse	0.68
401	XP 24000	0.62
402	SGX-SAS6-EM-Z Sun Storage 6 Gb SAS Express Module	0.46
403	PEG1 Single Port Copper Gigabit Ethernet PCI Express Server Adapter	0.34
404	SAS 9211-8i	0.33
405	HDC3-LPe	0.21

3. Nom i preu dels productes que costen més de 200€.

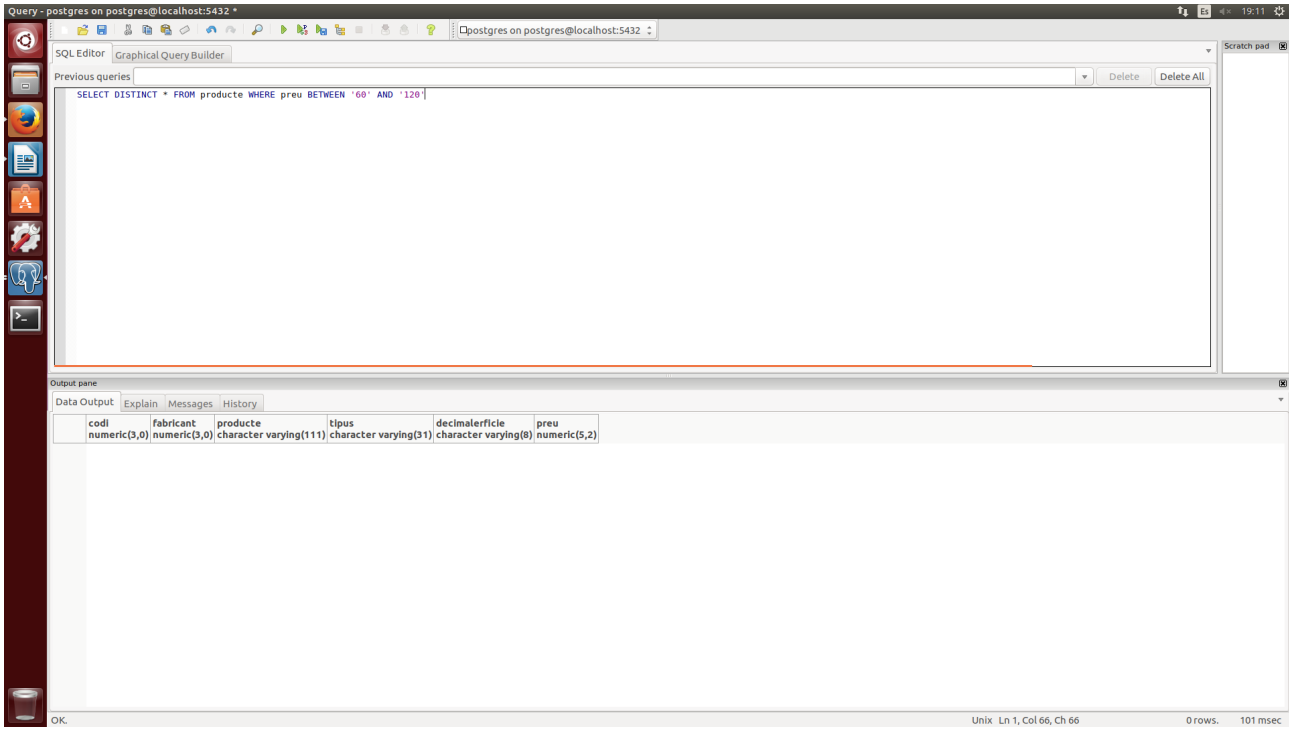
SELECT DISTINCT producte, preu FROM producte WHERE preu > ‘200’



Tasca 003 Llenguatge SQL.

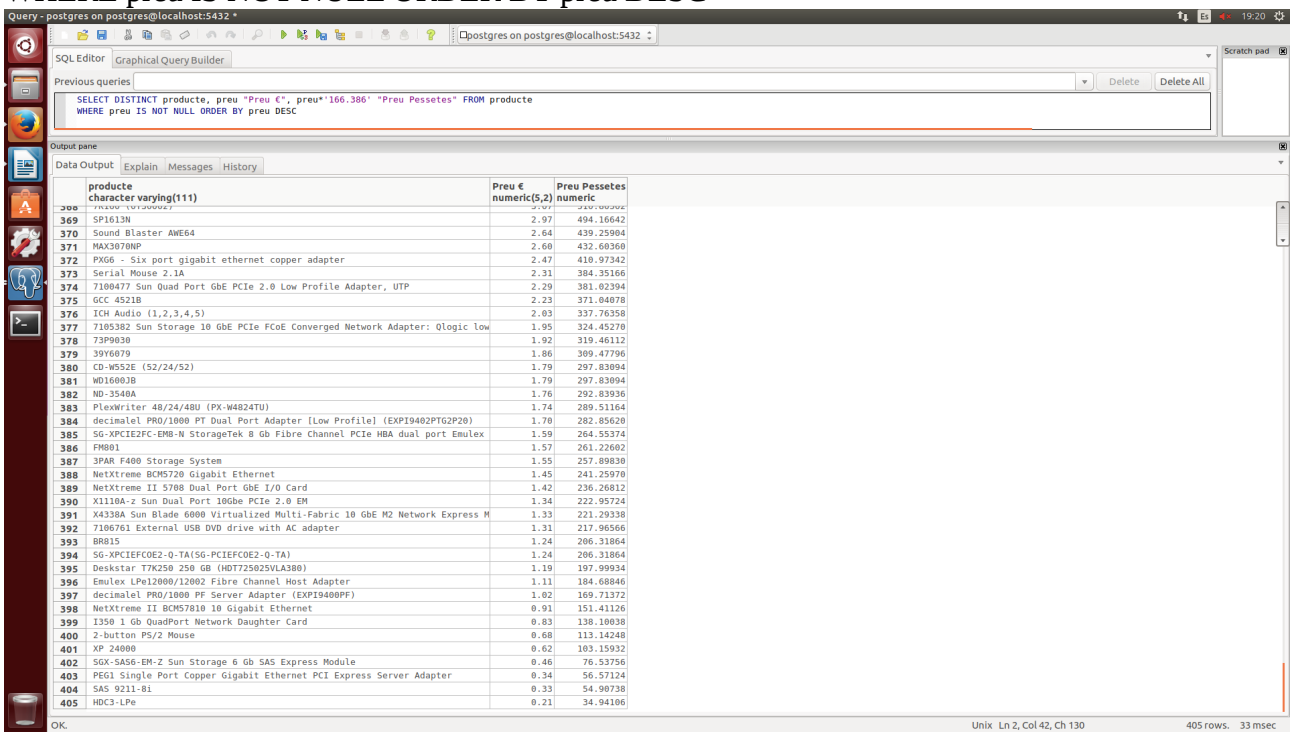
4. Obtindre totes les dades dels articles que tenen un preu entre 60€ i 120€. (fes ús del AND i del BETWEEN).

SELECT DISTINCT * FROM producte WHERE preu BETWEEN ‘60’ AND ‘120’



5. El nom i el preu en pessetes (preu en euros multiplicat per 166,386) dels articles de la botiga, fes ús d'un identificador per al preu en pessetes.

SELECT DISTINCT producte, preu "Preu €", preu*’166.386’ "Preu Pessetes" FROM producte WHERE preu IS NOT NULL ORDER BY preu DESC

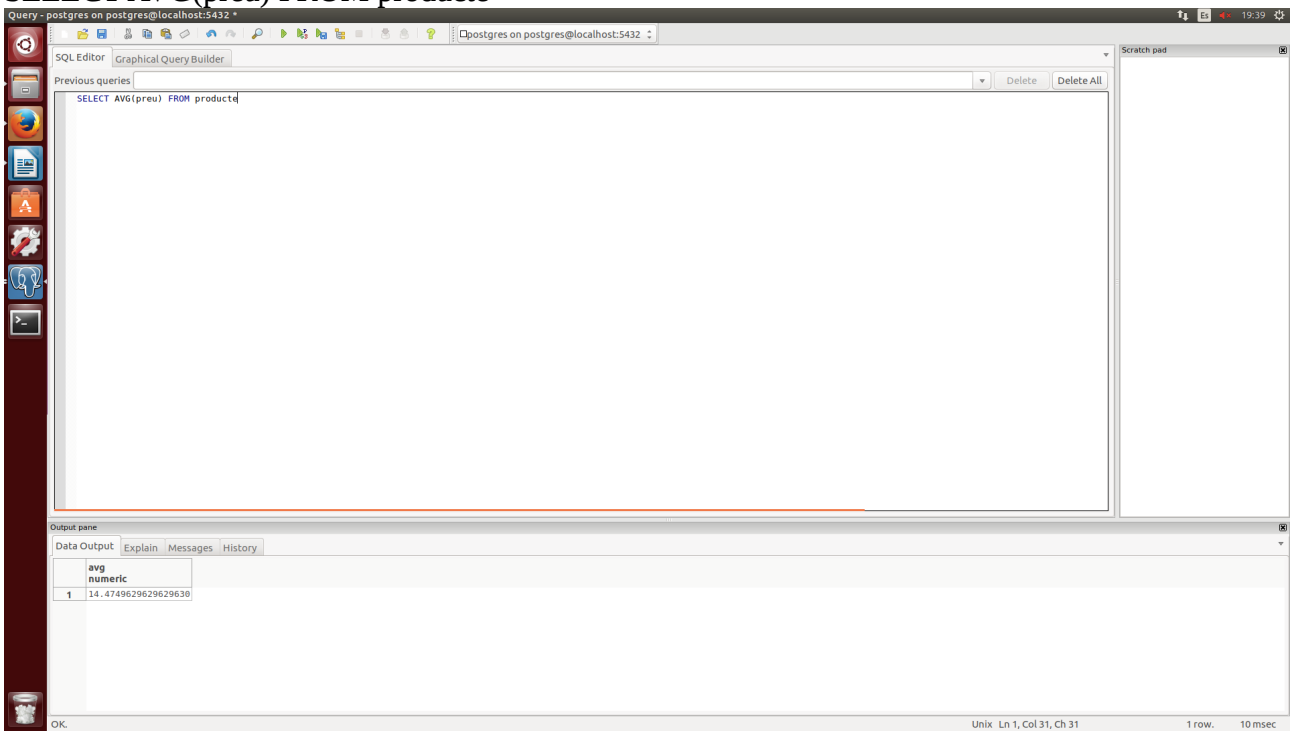


	producte character varying(111)	Preu € numeric(5,2)	Preu Pessetes numeric
369	SP1613N	2.97	494.16642
370	Sound Blaster AWE64	2.64	439.25904
371	MAX3070NP	2.60	432.60360
372	PXG6 - Six port gigabit ethernet copper adapter	2.47	410.97342
373	Serial Mouse 2.1A	2.31	384.35166
374	7100477 Sun Quad Port GbE PCIe 2.0 Low Profile Adapter, UTP	2.29	381.02394
375	GCC 4521B	2.23	371.04078
376	ICH Audio (1,2,3,4,5)	2.03	337.76358
377	7105382 Sun Storage 10 GbE PCIe FCoE Converged Network Adapter: Qlogic low	1.95	324.45270
378	73P9030	1.92	319.46112
379	39Y6079	1.86	309.47796
380	CD-W552E (52/24/52)	1.79	297.83094
381	WD1600JB	1.79	297.83094
382	ND-3540A	1.76	292.83936
383	PlexWriter 48/24/48U (PX-W4824TU)	1.74	289.51164
384	decimalel PRO/1000 PT Dual Port Adapter [Low Profile] (EXPI9402PTG2P20)	1.70	282.85620
385	SG-XPCIE2FC-EM8-N StorageTek 8 Gb Fibre Channel PCIe HBA dual port Emulex	1.59	264.55374
386	FM801	1.57	261.22602
387	3PAR F400 Storage System	1.55	257.89830
388	NetXtreme BCM5720 Gigabit Ethernet	1.45	241.25970
389	NetXtreme II 5708 Dual Port GbE I/O Card	1.42	236.26812
390	X1110A-z Sun Dual Port 10Gbe PCIe 2.0 EM	1.34	222.95724
391	X4338A Sun Blade 6000 Virtualized Multi-Fabric 10 GbE M2 Network Express M	1.33	221.29338
392	7106761 External USB DVD drive with AC adapter	1.31	217.96566
393	BR815	1.24	206.31864
394	SG-XPCIEFCOE2-Q-TA(SG-PCIEFCOE2-Q-TA)	1.24	206.31864
395	Deskstar T7K250 250 GB (HDT725025VLA380)	1.19	197.99934
396	Emulex LPe12000/12002 Fibre Channel Host Adapter	1.11	184.68846
397	decimalel PRO/1000 PF Server Adapter (EXPI9400PF)	1.02	169.71372
398	NetXtreme II BCM57810 10 Gigabit Ethernet	0.91	151.41126
399	I350 1 Gb QuadPort Network Daughter Card	0.83	138.10038
400	2-button PS/2 Mouse	0.68	113.14248
401	XP 24000	0.62	103.15932
402	SGX-SAS6-EM-Z Sun Storage 6 Gb SAS Express Module	0.46	76.53756
403	PEG1 Single Port Copper Gigabit Ethernet PCI Express Server Adapter	0.34	56.57124
404	SAS 9211-8i	0.33	54.90738
405	HDC3-LPe	0.21	34.94106

Tasca 003 Llenguatge SQL.

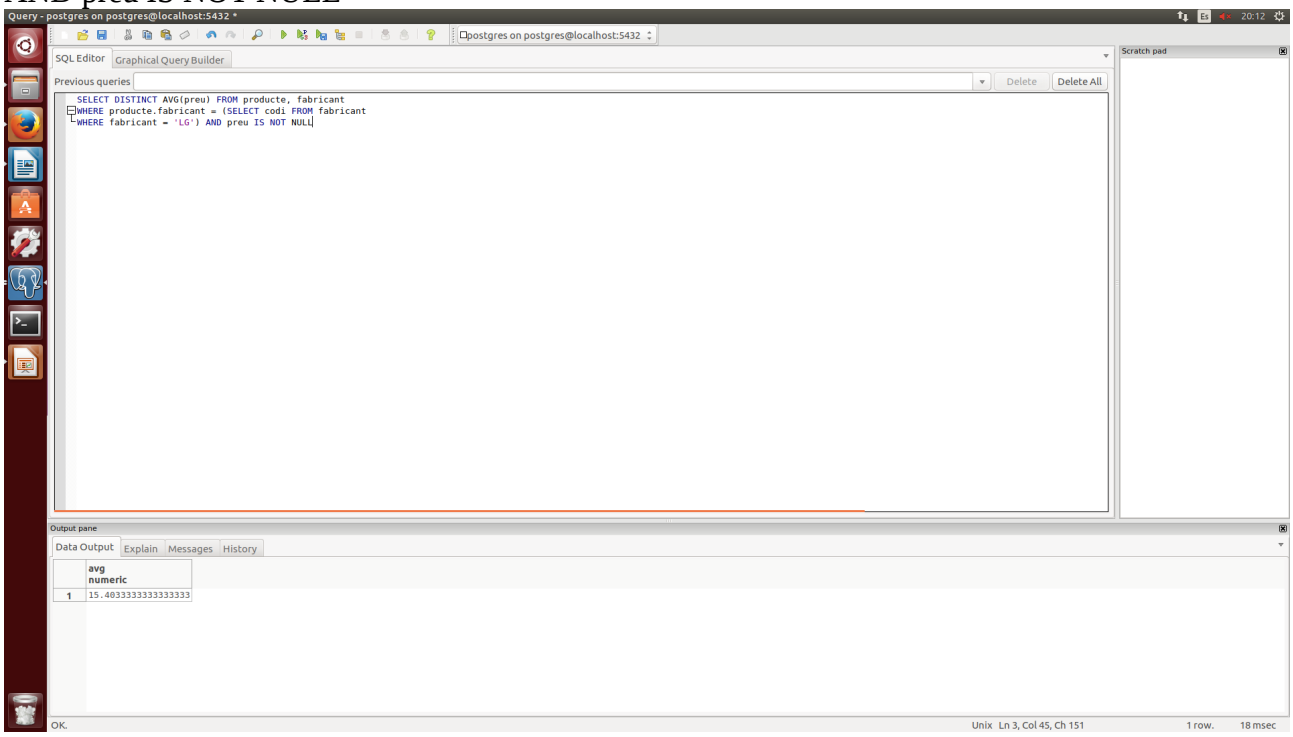
6. Seleccionar el preu mitjà de tots els articles.

SELECT AVG(preu) FROM producte



7. Calcula el preu mitjà de tots els articles que tenen com a fabricant LG.

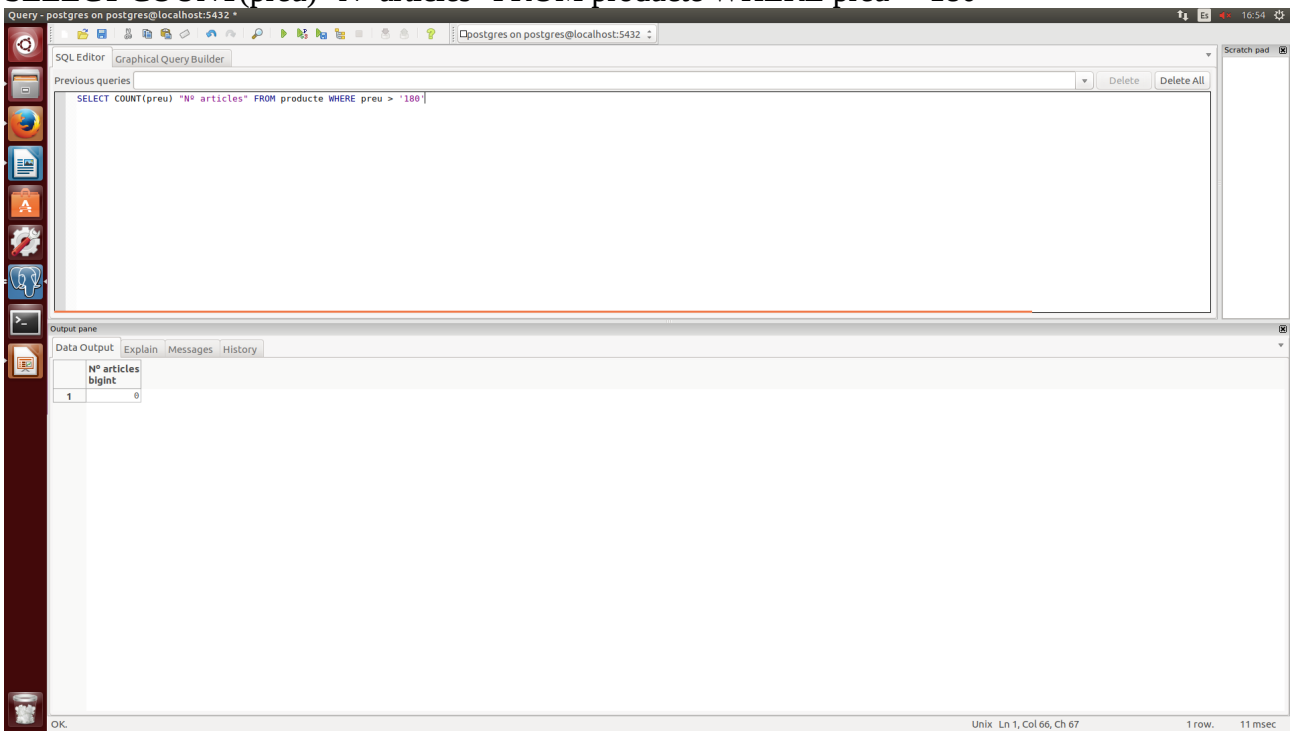
SELECT DISTINCT AVG(preu) FROM producte, fabricant
WHERE producte.fabricant = (SELECT codi FROM fabricant WHERE fabricant = ‘LG’)
AND preu IS NOT NULL



Tasca 003 Llenguatge SQL.

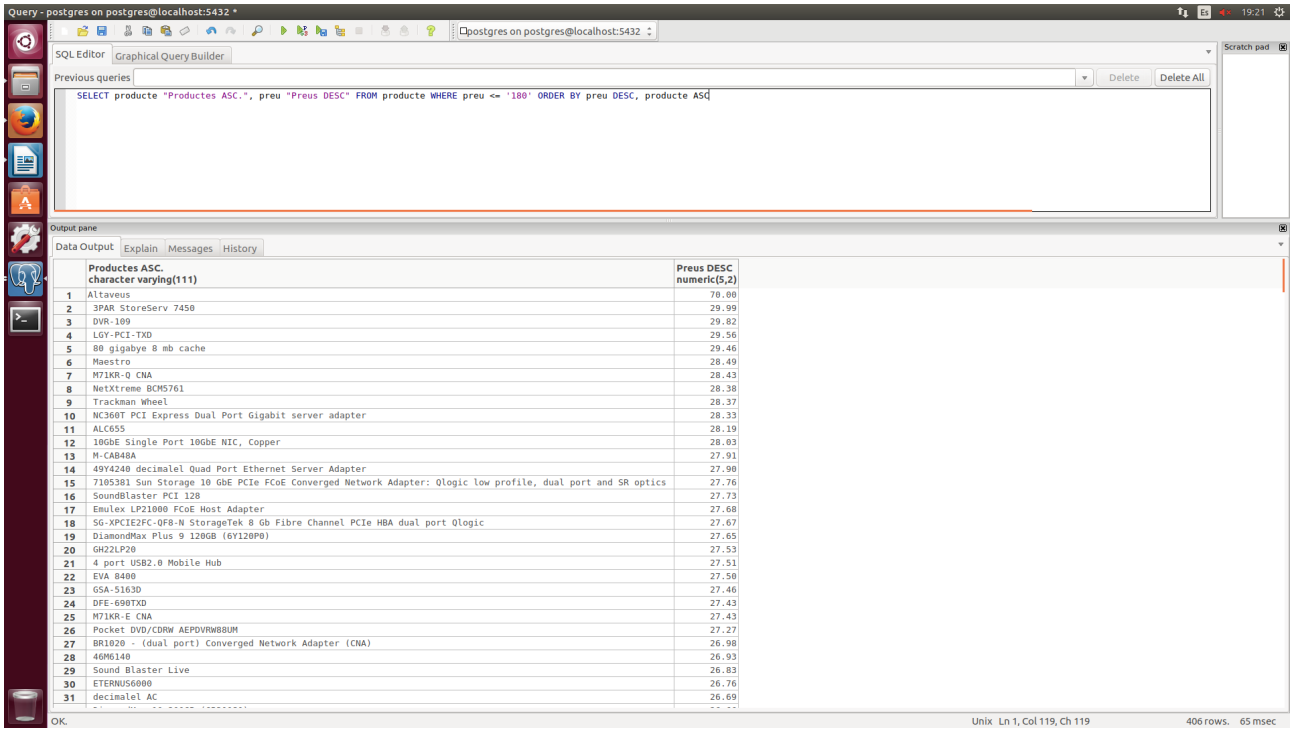
8. Número total d'articles que tenen un preu superior a 180€.

SELECT COUNT(preu) "Nº articles" FROM producte WHERE preu > '180'



9. Nom i preu dels articles que tenen un preu de 180€ o més ordenats descendentment pel preu, i ascendentment per nom.

SELECT producte "Productes ASC.", preu "Preus DESC" FROM producte WHERE preu <= '180' ORDER BY preu DESC, producte ASC

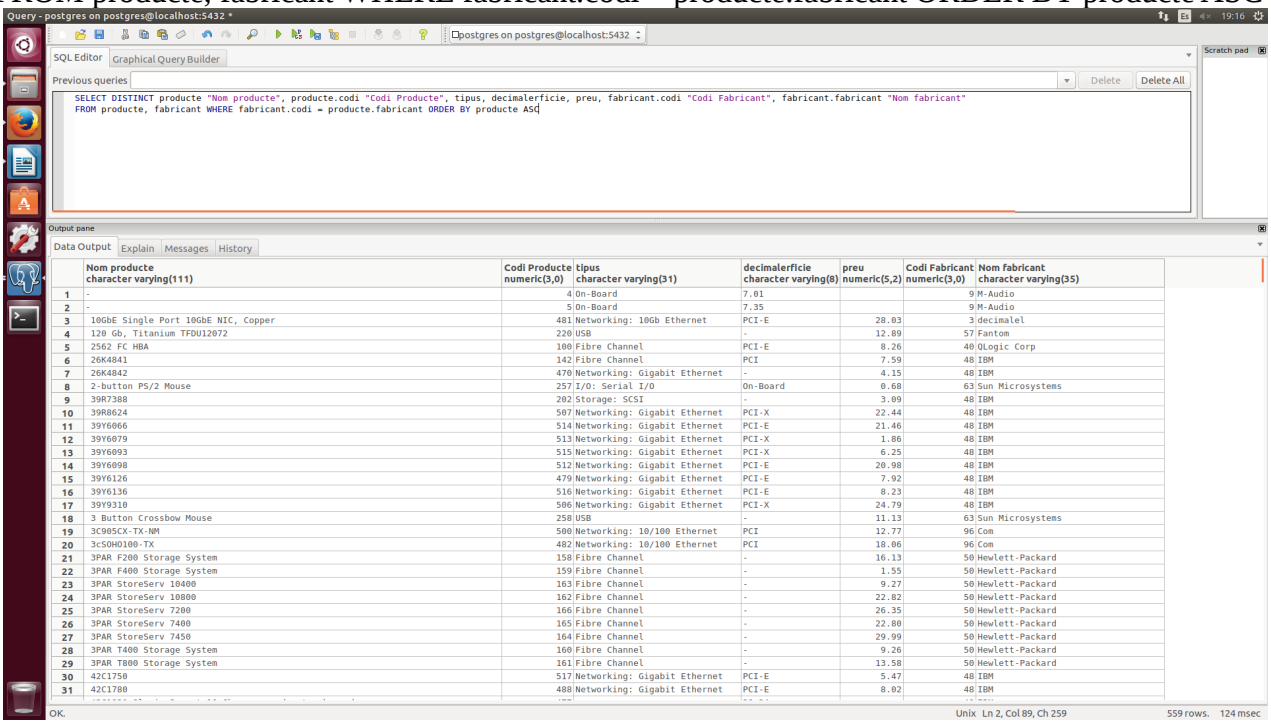


	Productes ASC. character varying(111)	Preus DESC numeric(5,2)
1	Altaveus	70.00
2	3PAR StoreServ 7450	29.99
3	DVR-109	29.82
4	LGY-PCI-TXD	29.56
5	80 gigabye 8 mb cache	29.46
6	Maestro	28.49
7	M71KR-Q CNA	28.43
8	NetXtreme BCM5761	28.38
9	Trackman Wheel	28.37
10	NC360T PCI Express Dual Port Gigabit server adapter	28.33
11	ALC655	28.19
12	10GbE Single Port 10GbE NIC, Copper	28.03
13	M-CAB48A	27.91
14	49Y4240 decimalel Quad Port Ethernet Server Adapter	27.90
15	7105381 Sun Storage 10 GbE PCIe FCoE Converged Network Adapter: Qlogic low profile, dual port and SR optics	27.76
16	SoundBlaster PCI 128	27.73
17	Emulex LP21000 FCoE Host Adapter	27.68
18	SG-XPCIE2FC-QF8-N StorageTek 8 Gb Fibre Channel PCIe HBA dual port Qlogic	27.67
19	DiamondMax Plus 9 120GB (6Y120P0)	27.65
20	GH22LP20	27.53
21	4 port USB2.0 Mobile Hub	27.51
22	EVA 8400	27.50
23	GSA-5163D	27.46
24	DFE-690TXD	27.43
25	M71KR-E CNA	27.43
26	Pocket DVD/CDRW AEPDVRWB8UM	27.27
27	BR1020 - (dual port) Converged Network Adapter (CNA)	26.98
28	46M6140	26.93
29	Sound Blaster Live	26.83
30	ETERNUS6000	26.76
31	decimalel AC	26.69

Tasca 003 Llenguatge SQL.

10. Obtindre un llistat complet d'articles, incloent per cada article les dades del article i del seu fabricant.

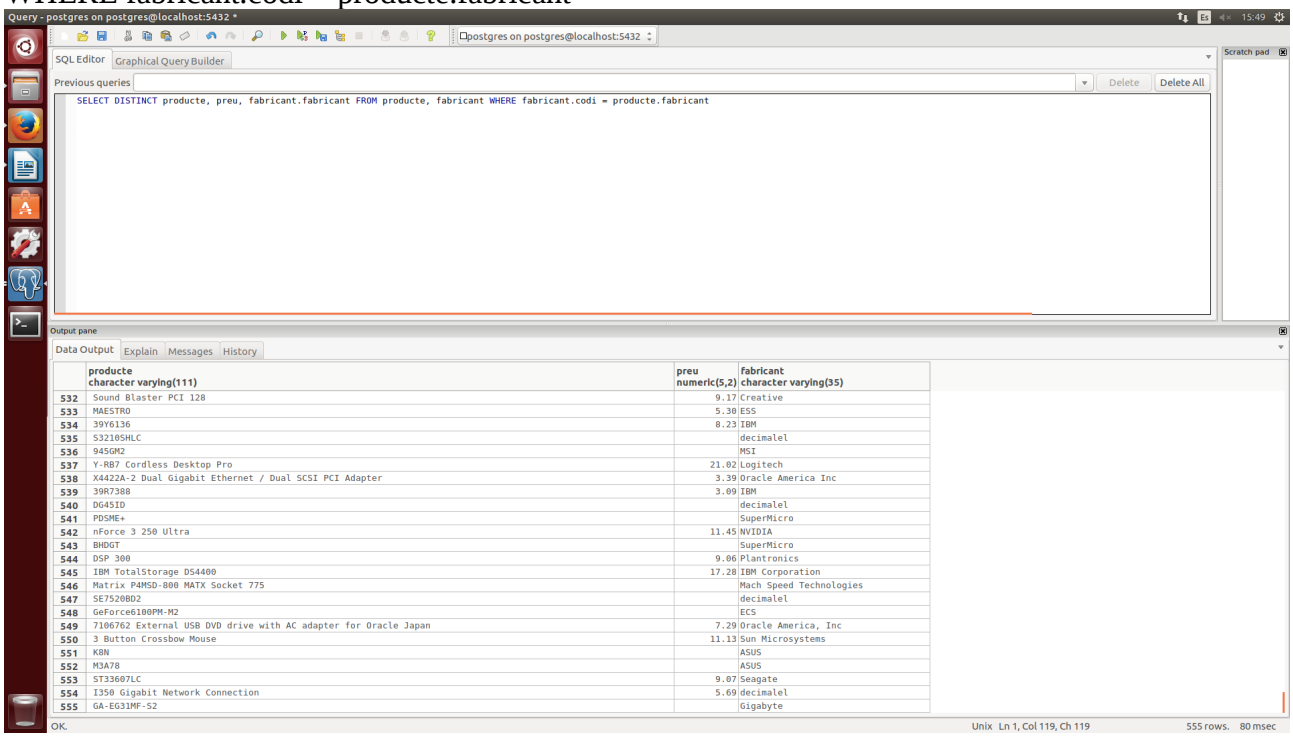
SELECT DISTINCT producte "Nom producte", producte.codi "Codi Producte", tipus, decimalerficie, preu, fabricant.codi "Codi Fabricant", fabricant.fabricant "Nom fabricant"
FROM producte, fabricant WHERE fabricant.codi = producte.fabricant ORDER BY producte ASC



	Nom producte character varying(111)	Codi Producte numeric(3,0)	tipus character varying(31)	decimalerficie character varying(8)	preu numeric(5,2)	Codi Fabricant numeric(3,0)	Nom fabricant character varying(35)
1	-	4	On-Board	7.01		9	M-Audio
2	-	5	On-Board	7.35		9	M-Audio
3	10GbE Single Port 10GbE NIC, Copper	481	Networking: 10Gb Ethernet	PCI-E	28.03	3	decimalel
4	120 Gb, Titanium TFDU12072	220	USB	-	12.89	57	Fantom
5	2562 FC HBA	100	Fibre Channel	PCI-E	8.26	40	QLogic Corp
6	26K4841	142	Fibre Channel	PCI	7.59	48	IBM
7	26K4842	470	Networking: Gigabit Ethernet	-	4.15	48	IBM
8	2-button PS/2 Mouse	257	I/O: Serial I/O	On-Board	0.68	63	Sun Microsystems
9	39R7388	202	Storage: SCSI	-	3.09	48	IBM
10	39R8624	507	Networking: Gigabit Ethernet	PCI-X	22.44	48	IBM
11	39Y6066	514	Networking: Gigabit Ethernet	PCI-E	21.46	48	IBM
12	39Y6079	513	Networking: Gigabit Ethernet	PCI-X	1.86	48	IBM
13	39Y6093	515	Networking: Gigabit Ethernet	PCI-X	6.25	48	IBM
14	39Y6098	512	Networking: Gigabit Ethernet	PCI-E	20.98	48	IBM
15	39Y6126	479	Networking: Gigabit Ethernet	PCI-E	7.92	48	IBM
16	39Y6136	516	Networking: Gigabit Ethernet	PCI-E	8.23	48	IBM
17	39Y9310	506	Networking: Gigabit Ethernet	PCI-X	24.79	48	IBM
18	3 Button Crossbow Mouse	258	USB	-	11.13	63	Sun Microsystems
19	3C905CX-TX-NM	500	Networking: 10/100 Ethernet	PCI	12.77	96	Com
20	3cSOHO100-TX	482	Networking: 10/100 Ethernet	PCI	18.06	96	Com
21	3PAR F200 Storage System	150	Fibre Channel	-	16.13	50	Hewlett-Packard
22	3PAR F400 Storage System	159	Fibre Channel	-	1.55	50	Hewlett-Packard
23	3PAR StoreServ 10400	163	Fibre Channel	-	9.27	50	Hewlett-Packard
24	3PAR StoreServ 10800	162	Fibre Channel	-	22.82	50	Hewlett-Packard
25	3PAR StoreServ 7200	166	Fibre Channel	-	26.35	50	Hewlett-Packard
26	3PAR StoreServ 7400	165	Fibre Channel	-	22.80	50	Hewlett-Packard
27	3PAR StoreServ 7450	164	Fibre Channel	-	29.99	50	Hewlett-Packard
28	3PAR T400 Storage System	160	Fibre Channel	-	9.26	50	Hewlett-Packard
29	3PAR T800 Storage System	161	Fibre Channel	-	13.58	50	Hewlett-Packard
30	42C1750	517	Networking: Gigabit Ethernet	PCI-E	5.47	48	IBM
31	42C1780	488	Networking: Gigabit Ethernet	PCI-E	8.02	48	IBM

11. Obtindre un llistat d'articles, incloent el nom de l'article i el seu preu amb el nom del fabricant.

SELECT producte, preu, fabricant.fabricant FROM producte, fabricant
WHERE fabricant.codi = producte.fabricant

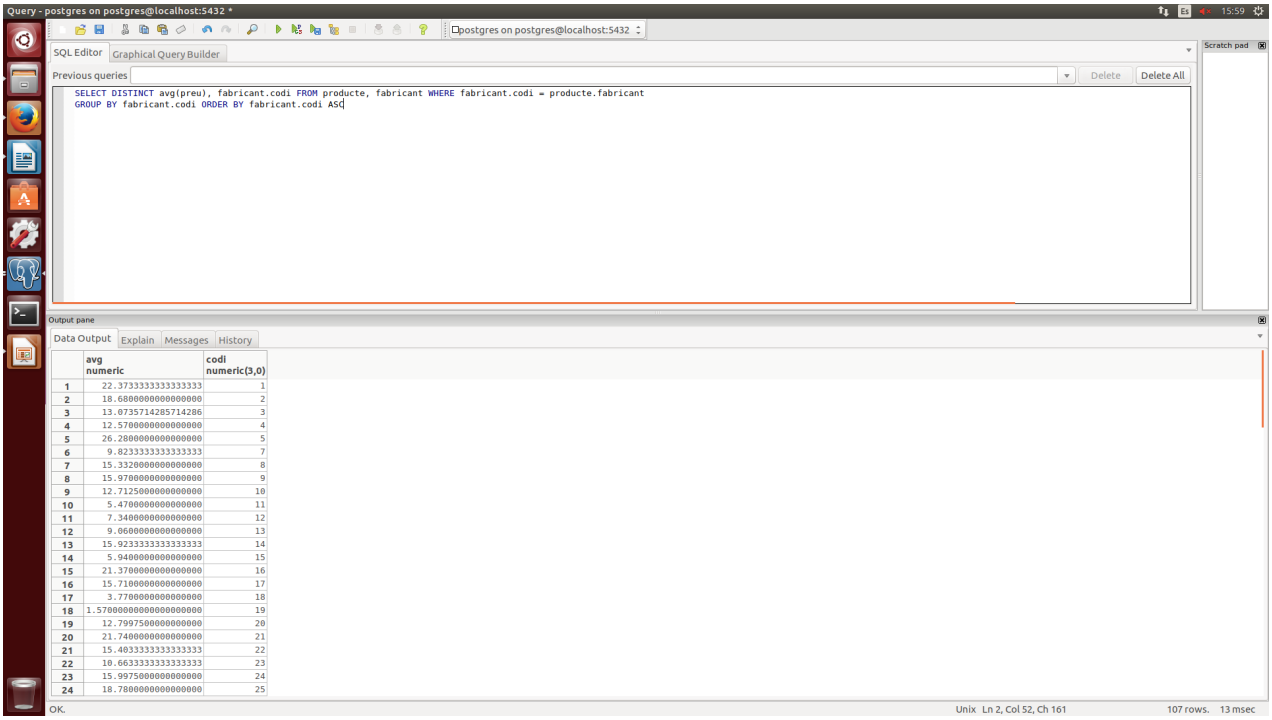


	producte character varying(111)	preu numeric(5,2)	fabricant character varying(35)
532	Sound Blaster PCI 128	9.17	Creative
533	MAESTRO	5.30	ESS
534	39Y6136	8.23	IBM
535	S3210SHLC		decimalel
536	945GM2		MSI
537	Y-RB7 Cordless Desktop Pro	21.02	Logitech
538	X4422A-2 Dual Gigabit Ethernet / Dual SCSI PCI Adapter	3.39	Oracle America Inc
539	39R7388	3.09	IBM
540	DG45ID		decimalel
541	PD5ME+		SuperMicro
542	nForce 3 250 Ultra	11.45	NVIDIA
543	BHDGT		SuperMicro
544	DSP 300	9.06	Plantronics
545	IBM TotalStorage DS4400	17.28	IBM Corporation
546	Matrix P4MSD-800 MATX Socket 775		Mach Speed Technologies
547	SE7520BD2		decimalel
548	GeForce6100PM-M2		ECS
549	7106762 External USB DVD drive with AC adapter for Oracle Japan	7.29	Oracle America, Inc
550	3 Button Crossbow Mouse	11.13	Sun Microsystems
551	K8N		ASUS
552	M3A78		ASUS
553	ST33607LC	9.07	Seagate
554	I350 Gigabit Network Connection	5.69	decimalel
555	GA-EG31MF-S2		Gigabyte

Tasca 003 Llenguatge SQL.

12. Obtindre el preu mig dels productes de cada fabricant, mostrant el codi de fabricant.

SELECT DISTINCT avg(preu), fabricant.codi FROM producte, fabricant
WHERE fabricant.codi = producte.fabricant GROUP BY fabricant.codi
ORDER BY fabricant.codi ASC

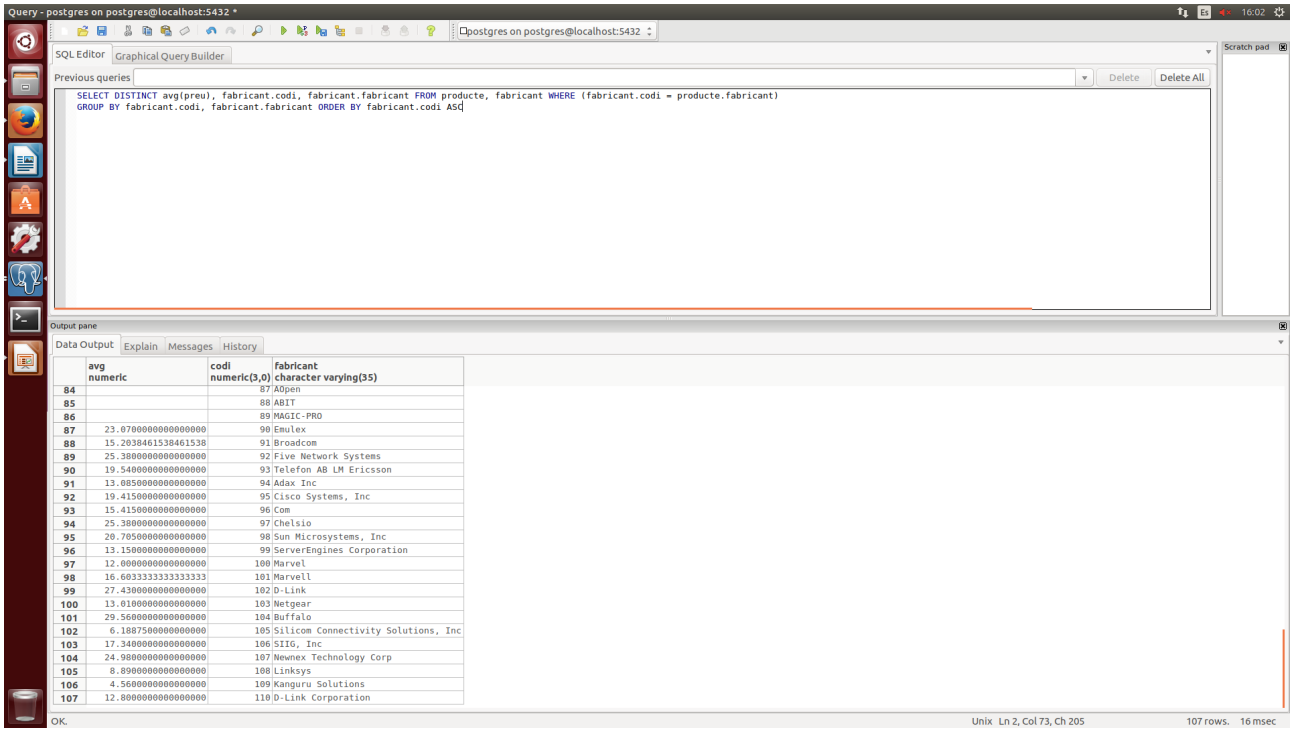


	avg numeric	codi numeric(3,0)
1	22.3733333333333333	1
2	18.6800000000000000	2
3	13.0735714285714286	3
4	12.5700000000000000	4
5	26.2800000000000000	5
6	9.8233333333333333	7
7	15.3320000000000000	8
8	15.9700000000000000	9
9	12.7125000000000000	10
10	5.4700000000000000	11
11	7.3400000000000000	12
12	9.0600000000000000	13
13	15.9233333333333333	14
14	5.9400000000000000	15
15	21.3700000000000000	16
16	15.7100000000000000	17
17	3.7700000000000000	18
18	1.5700000000000000000	19
19	12.7997500000000000	20
20	21.7400000000000000	21
21	15.4033333333333333	22
22	10.6633333333333333	23
23	15.9975000000000000	24
24	18.7800000000000000	25

Tasca 003 Llenguatge SQL.

13. Obtindre el preu mig dels productes de cada fabricant, mostrant el codi i el seu nom de fabricant.

SELECT DISTINCT avg(preu), fabricant.codi, fabricant.fabricant FROM producte, fabricant WHERE (fabricant.codi = producte.fabricant) GROUP BY fabricant.codi, fabricant.fabricant ORDER BY fabricant.codi ASC

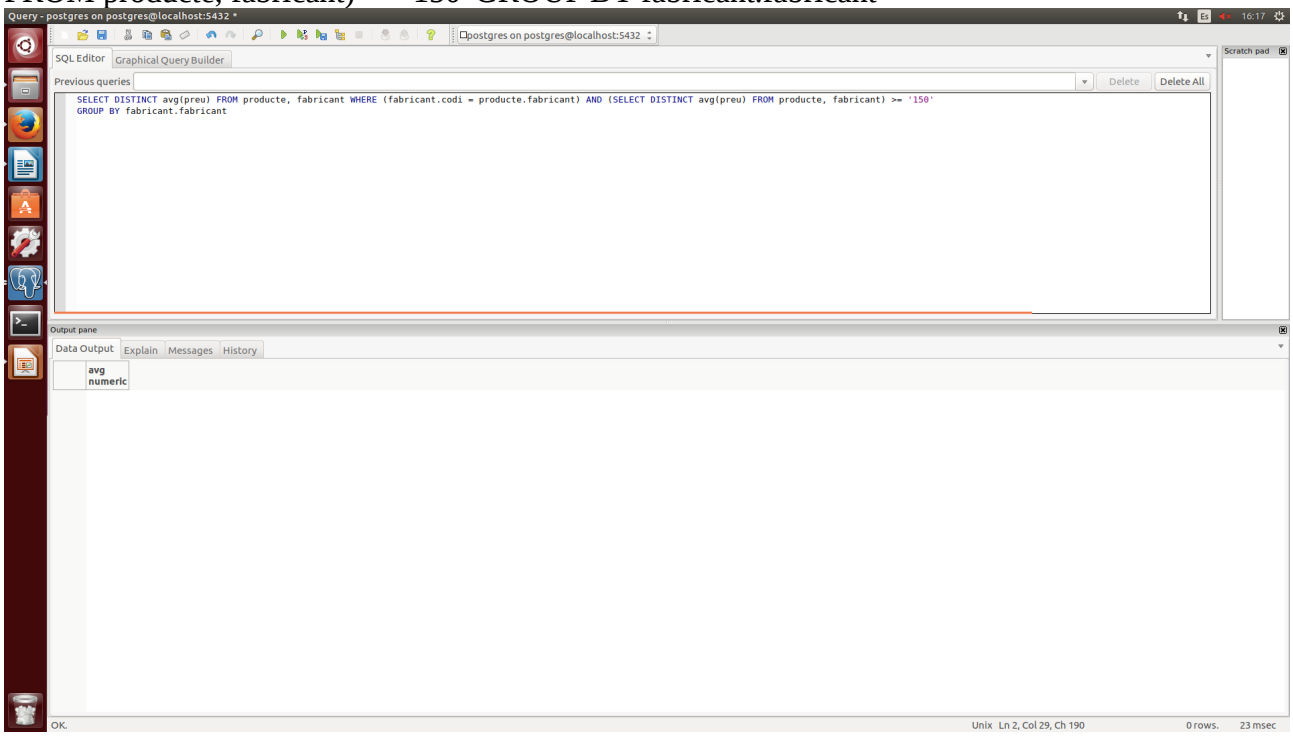


	avg numeric	codi numeric(3,0)	fabricant character varying(35)
84		87	AOpen
85		88	ABIT
86		89	MAGIC-PRO
87	23.0700000000000000	90	Emulex
88	15.2038461538461538	91	Broadcom
89	25.3800000000000000	92	Five Network Systems
90	19.5400000000000000	93	Telefon AB LM Ericsson
91	13.0850000000000000	94	Adax Inc
92	19.4150000000000000	95	Cisco Systems, Inc
93	15.4150000000000000	96	Com
94	25.3800000000000000	97	Chelsio
95	20.7050000000000000	98	Sun Microsystems, Inc
96	13.1500000000000000	99	ServerEngines Corporation
97	12.0000000000000000	100	Marvel
98	16.6033333333333333	101	Marvell
99	27.4300000000000000	102	D-Link
100	13.0100000000000000	103	Netgear
101	29.5600000000000000	104	Buffalo
102	6.1887500000000000	105	Silicom Connectivity Solutions, Inc
103	17.3400000000000000	106	SIIG, Inc
104	24.9800000000000000	107	Newnex Technology Corp
105	8.8900000000000000	108	Linksys
106	4.5600000000000000	109	Kanguru Solutions
107	12.8000000000000000	110	D-Link Corporation

Tasca 003 Llenguatge SQL.

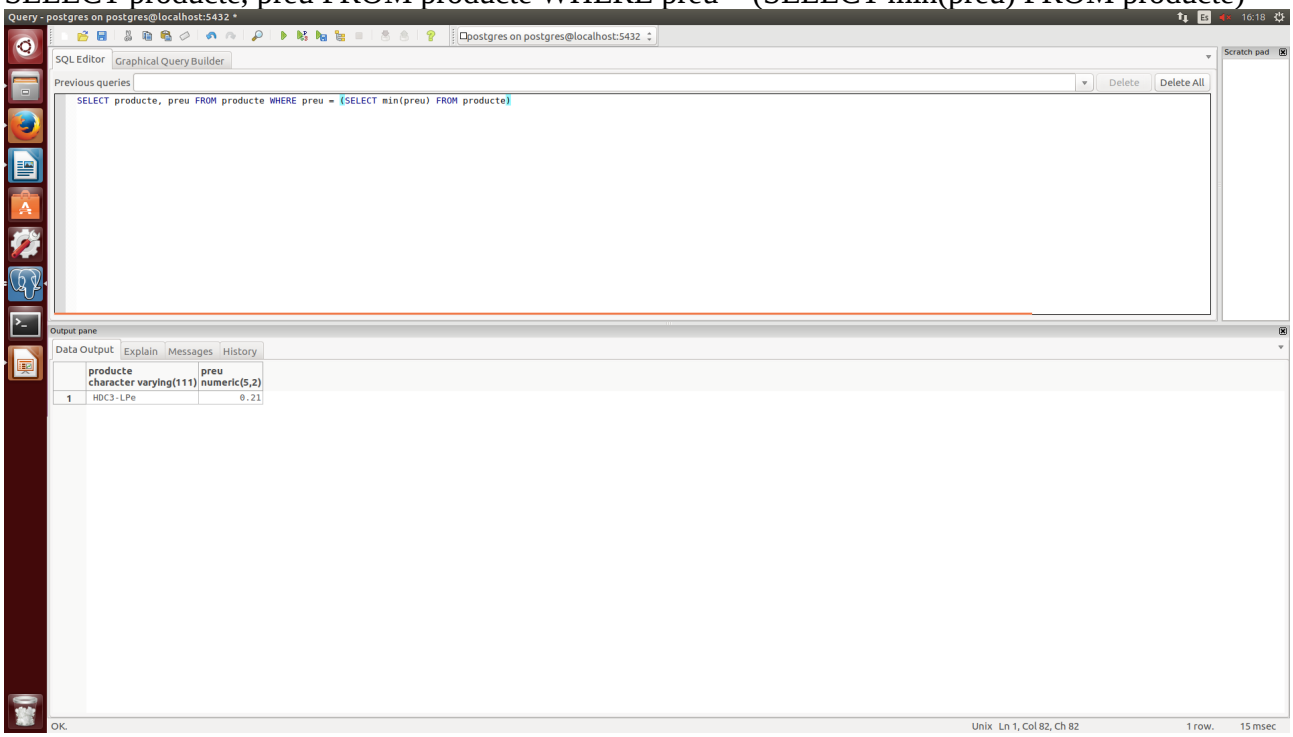
14. Obtindre els noms dels fabricants que ofereixen productes amb un preu mig major o igual a 150€.

SELECT DISTINCT avg(preu) FROM producte, fabricant
WHERE (fabricant.codi = producte.fabricant) AND (SELECT DISTINCT avg(preu)
FROM producte, fabricant) >= '150' GROUP BY fabricant.fabricant



15. Obtindre el nom i el preu de l'article més barat.

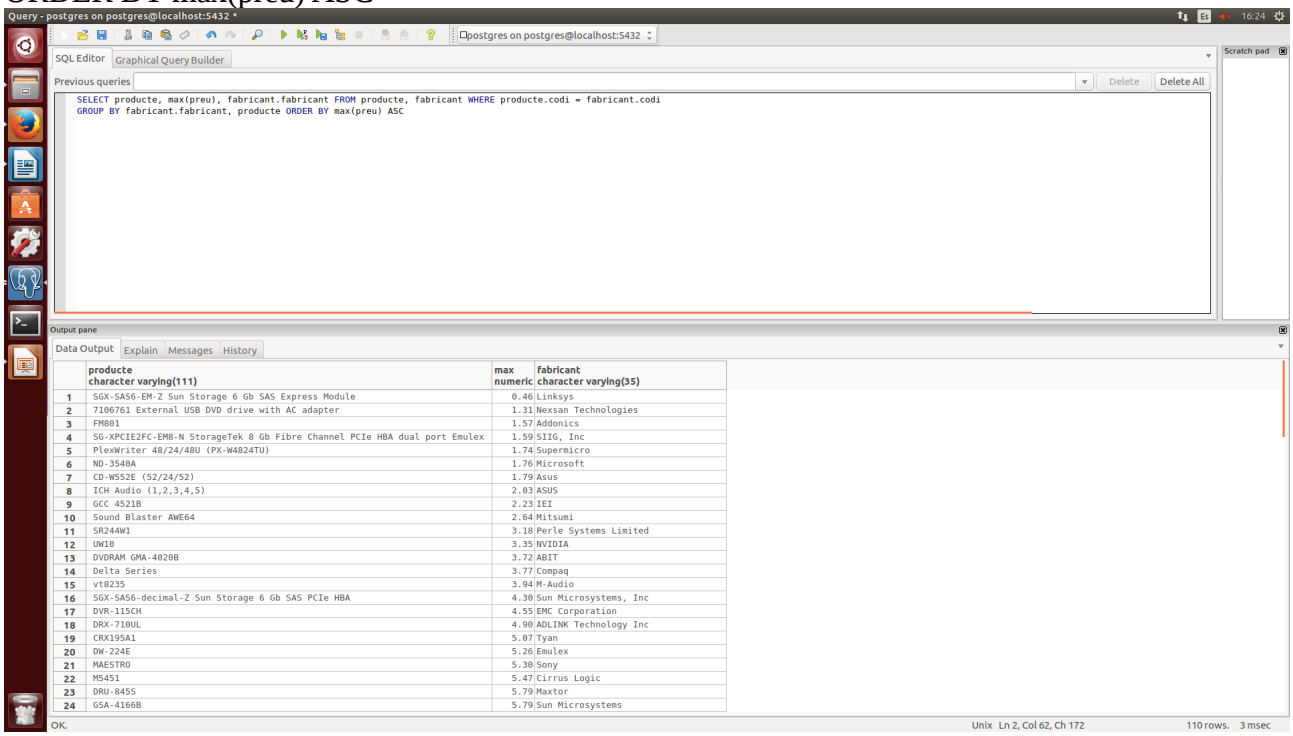
SELECT producte, preu FROM producte WHERE preu = (SELECT min(preu) FROM producte)



Tasca 003 Llenguatge SQL.

16. Obtindre una llista amb el nom i el preu dels articles mes cars de cada proveïdor (incloent el nom del proveïdor).

SELECT producte, max(preu), fabricant.fabricant FROM producte, fabricant
WHERE producte.codi = fabricant.codi GROUP BY fabricant.fabricant, producte
ORDER BY max(preu) ASC



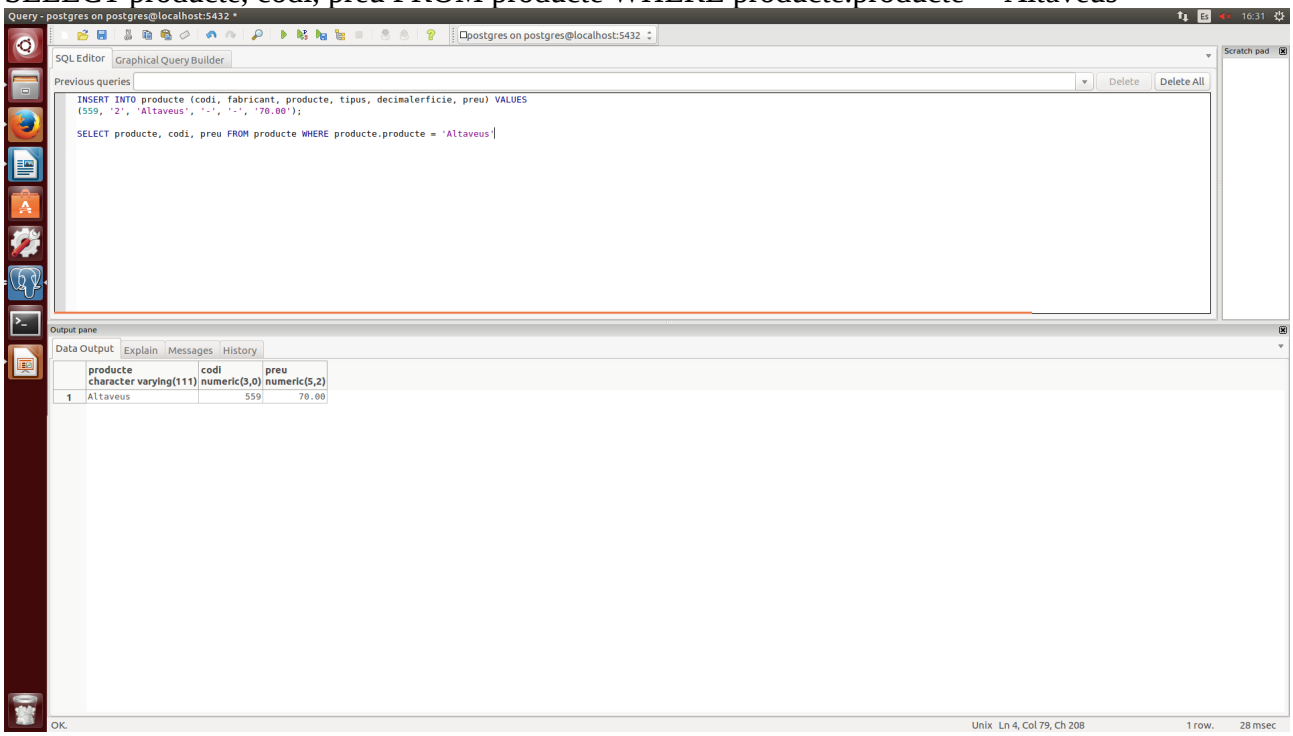
	producte character varying(111)	max numeric	fabricant character varying(35)
1	SGX-SAS6-EM-Z Sun Storage 6 Gb SAS Express Module	0.46	Linksys
2	7106761 External USB DVD drive with AC adapter	1.31	Nexsan Technologies
3	FM801	1.57	Addonics
4	SG-XPCIE2FC-EM8-N StorageTek 8 Gb Fibre Channel PCIe HBA dual port Emulex	1.59	SIIG, Inc
5	PlexWriter 48/24/48U (PX-W4824TU)	1.74	Supermicro
6	ND-3540A	1.76	Microsoft
7	CD-W552E (52/24/52)	1.79	Asus
8	ICH Audio (1,2,3,4,5)	2.03	ASUS
9	GCC 4521B	2.23	IEI
10	Sound Blaster AWE64	2.64	Mitsumi
11	SR244W1	3.18	Perle Systems Limited
12	UW10	3.35	NVIDIA
13	DVDRAM GMA-4020B	3.72	ABIT
14	Delta Series	3.77	Compaq
15	vt8235	3.94	M-Audio
16	SGX-SAS6-decimal-Z Sun Storage 6 Gb SAS PCIe HBA	4.30	Sun Microsystems, Inc
17	DVR-115CH	4.55	EMC Corporation
18	DRX-710UL	4.90	ADLINK Technology Inc
19	CRX195A1	5.07	Tyan
20	DW-224E	5.26	Emulex
21	MAESTRO	5.30	Sony
22	M5451	5.47	Cirrus Logic
23	DRU-845S	5.79	Maxtor
24	GSA-4166B	5.79	Sun Microsystems

Tasca 003 Llenguatge SQL.

17. Afegir un nou producte altaveus de 70€ (del fabricant 2).

INSERT INTO producte (codi, fabricant, producte, tipus, decimalerficie, preu) VALUES (559, '2', 'Altaveus', '-', '-', '70.00');

SELECT producte, codi, preu FROM producte WHERE producte.producte = 'Altaveus'

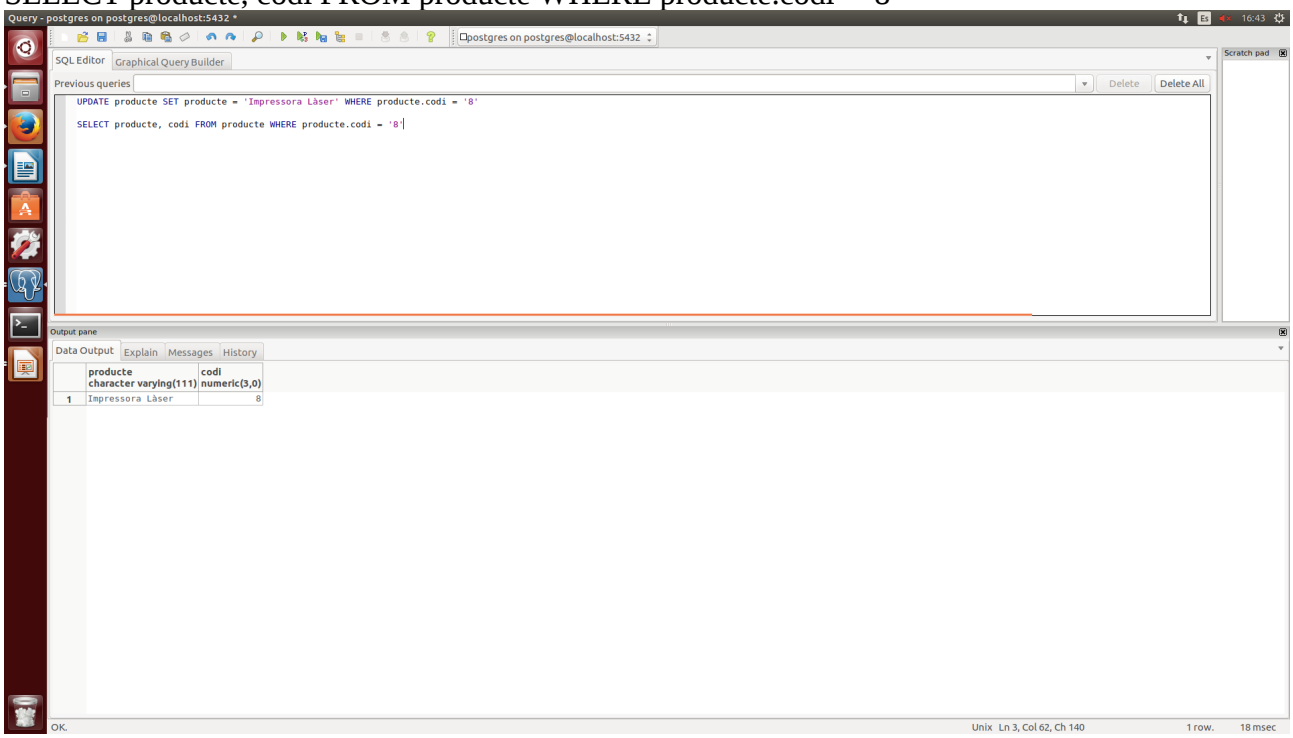


Tasca 003 Llenguatge SQL.

18. Canvia el nom del producte 8 a impressora Laser.

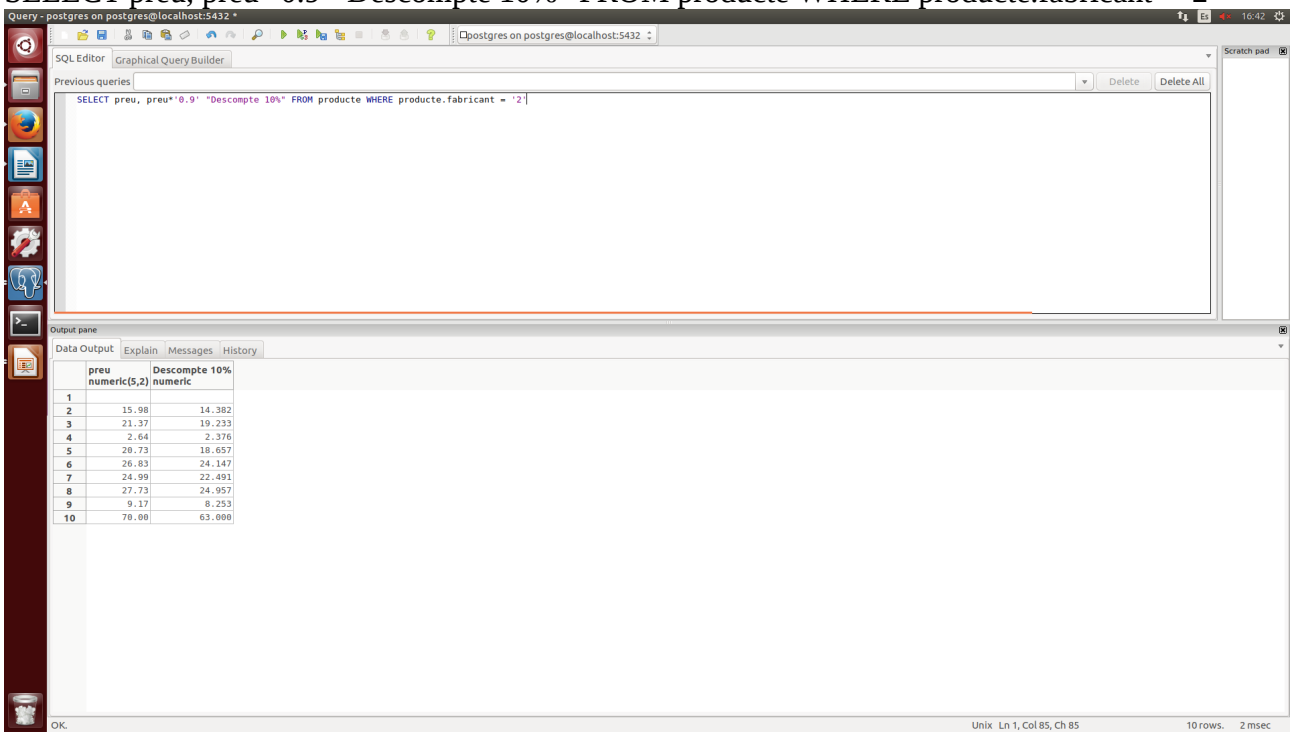
UPDATE producte SET producte = 'Impressora Làser' WHERE producte.codi = '8'

SELECT producte, codi FROM producte WHERE producte.codi = '8'

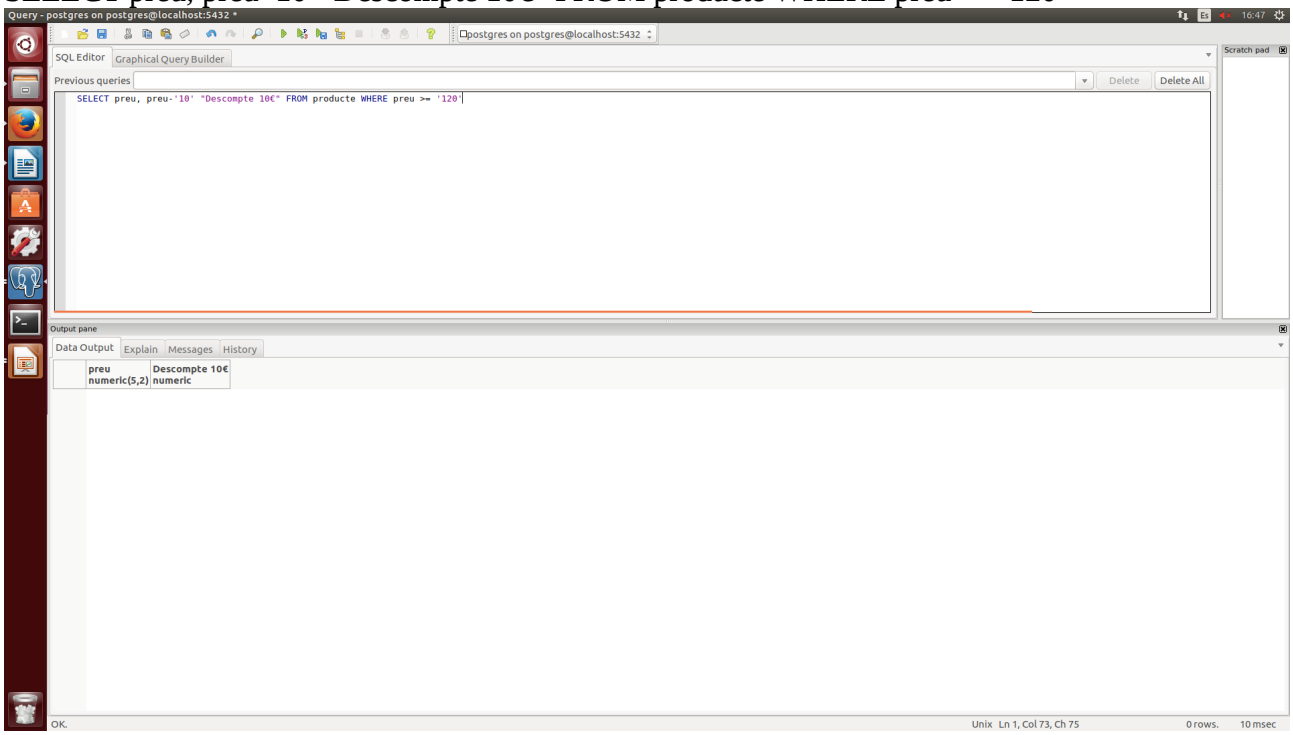


19. Aplica un descompte del 10% (multiplicar preu per 0,9) a tots els productes del fabricant 2.

SELECT preu, preu*'0.9' "Descompte 10%" FROM producte WHERE producte.fabricant = '2'



Tasca 003 Llenguatge SQL.

20. Aplicar un descompte de 10 euros a tots els productes que tinguen un preu major o igual a 120€.

SELECT preu, preu-'10' "Descompte 10€" FROM producte WHERE preu >= '120'